PHRASEBOOK

The Language of Maths

for people who were never taught the words

$2 + 3 = 5$ ⟷ *two add three is five*

Before any maths: a promise

If maths has always looked like a wall of strange marks, that is **not** because you are bad at it. Maths is written in a language, a very squashed-up, shorthand language, and almost every book expects you to speak it already. Nobody ever sat you down and taught you the words.

This part does exactly that. Think of it as a **phrasebook** for a country you are about to visit. You do not need to be fluent. You only need to be able to read the signs, and to know what to say when someone points at one.

There are no sums to do in this part. There is only reading.

How the rest of this book works

- Each section starts with a boxed RULE (or a numbered PROCEDURE) that says exactly what to do, then worked examples, then a picture only if it helps. Every chapter ends with a rules page listing all its rules, and the Formula Pages at the back collect them all.
- In worked examples, each line changes only one thing. The part that changed is marked, and a handwritten note says why you are allowed to do it.
- Before you are asked to trust a rule, it is tested with real numbers so you can watch it work.
- This is a Xournal++ file, so write on it. Circle what you don't get. Highlight what you do.
- Take it slowly. Ten minutes on one line is normal. Mathematicians do it too.

Lesson 1 · Maths is shorthand

Here is a sentence in ordinary English. Read it slowly. Then follow the arrows: each coloured chunk gets turned into maths shorthand.

I think of a number, → x (x is a letter standing for the mystery number)

double it, → $2x$ (2x means 2 times x)

add three, → $+3$ (add 3)

and I get eleven. → $= 11$ (is the same value as 11)

Put the pieces together and the whole sentence becomes:

$$2x + 3 = 11$$

That is all an equation is: a sentence, written in shorthand. The maths version is shorter, but it says **exactly the same thing** as the English. Every time you see maths, there is an English sentence hiding inside it. Your job is only to find that sentence.

It works the other way too. Here is a piece of maths you might see in a shop:

$$C = p \times n$$

C — the total cost

p — the price of one

n — how many you buy

So $C = p \times n$ is read out as: "the total cost is the price of one thing times the number of things you buy". The letters are just the first letters of the words.

Why bother with shorthand? Because the long English sentences get very tiring very quickly, and because the short version is easier to move around. Later in this book you will learn to shuffle shorthand sentences around to find missing numbers. You cannot do that easily with paragraphs of English.

Lesson 2 · Doing things to numbers

Every line below is one phrase. The English is on the left, the **maths** is on the right, and underneath is how people actually *say it out loud*.

Put 3 and 4 together. $3 + 4$

say: three plus four

Start with 7 and take 2 away. $7 - 2$

say: seven minus two

Three lots of four (4 + 4 + 4). 3×4

say: three times four

The times sign has disguises. All of these mean 3 times 4: 3×4, $3 \cdot 4$, $3 * 4$ (on computers), $3(4)$ and $(3)(4)$.

Share 12 equally into 3 groups. $12 \div 3$

say: twelve divided by three

Division also has disguises: $12 \div 3$, $12/3$ and $\frac{12}{3}$. The line in a fraction **is** a division sign.

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5 multiplied by itself (5 times 5). 5^2

say: five squared

The small raised 2 means "multiply two copies together": $5^2 = 5 \times 5 = 25$. It is **not** 5 times 2. (All of Chapter 9 is about these.)

Which number, times itself, makes 9? $\sqrt{9}$

say: the square root of nine

The answer is 3, because $3 \times 3 = 9$. So $\sqrt{9} = 3$. A square root **undoes** squaring.

Four below zero (like a temperature of minus 4 degrees). -4

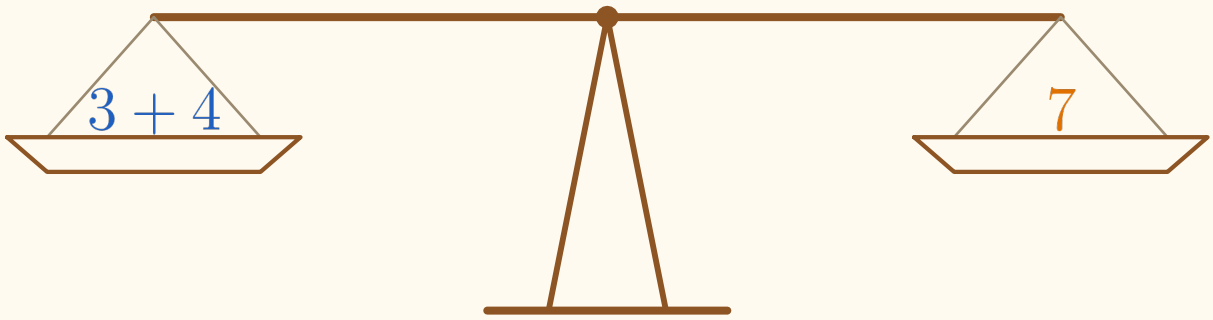
say: minus four / negative four

Watch out: the letter x and the times sign $\times$ look almost the same. That is why, once letters appear, people usually leave the times sign out. You will see this on the next pages.

Lesson 3 · The = sign is a balance

Most people learn at school that "=" means "and the answer is...". That idea causes a huge amount of confusion later on. Here is what it really means:

= means **"the thing on the left has exactly the same value as the thing on the right"**. It is a **balanced see-saw**.



same value on both sides, so it balances

Because it is a balance, it does not matter which side is which:

$3 + 4 = 7$ ✓ $\quad 7 = 3 + 4$ ✓ $\quad 3 + 4 = 2 + 5$ ✓

All three are true. The last one surprises people: there is no "answer" on the right at all, just two different ways of making 7. They balance, so the = sign is correct.

Why this matters later: when you rearrange formulas (Chapter 14) and solve equations (Chapter 15), the whole trick is to keep the see-saw balanced. Whatever you do to one side, you must do to the other side too.

The rest of the balance family

not the same value ..... $3 \neq 5$

say: three is not equal to five

smaller than ..... $3 < 8$

say: three is less than eight

bigger than ..... $8 > 3$

say: eight is greater than three

smaller than, or maybe equal to ..... $x \leq 10$

say: x is less than or equal to ten

bigger than, or maybe equal to ..... $x \geq 0$

say: x is greater than or equal to zero

roughly the same ..... $\pi \approx 3.14$

say: pi is approximately three point one four

A trick for < and >

The sign is a mouth. It always opens wide towards the bigger number and points its narrow tip at the smaller one. (It is "greedy": it wants to eat the bigger number.)



wide end at the 8

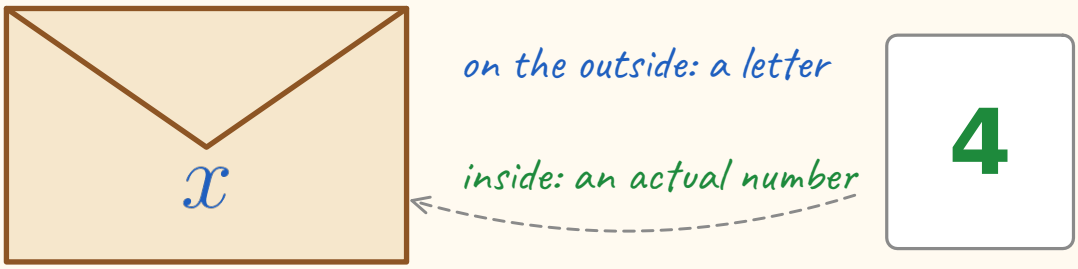
wide end at the 8

Your Lecture 1 slides also use three rarer members of the family:

- $1000000 \gg 2$ *say: is much greater than.* Far, far bigger.
- $0.001 \ll 5$ *say: is much less than.* Far, far smaller.
- $a + b \equiv b + a$ *say: is identically equal to.* Three lines instead of two means "this is not just true for one number, it is **always** true, whatever numbers you use".

This is the step that loses most people, so let's go very slowly.

A **letter** in maths is **a number wearing a disguise**. Picture a sealed envelope with a number written on a card inside. We cannot see the number, so we write a letter on the outside of the envelope to talk about it.



Letters do one of two jobs:

1. A mystery number. In $2x + 3 = 11$ there is exactly one number hiding in the envelope. (It's 4: double 4 is 8, add 3 is 11.) Finding it is called solving.

2. A number that can change. In $C = p \times n$ the letters change from shop to shop and day to day. The shorthand is a recipe that works whatever the numbers are. A letter used like this is called a variable (because it can vary).

How letters are written (the rules nobody tells you)

3 times x ---------- $3x$

say: three x

The times sign is left out. The number is written **first**: $3x$, never $x3$.

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x times y ---------- xy

say: x y

Two letters side by side are multiplied.

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x times x ---------- x^2

say: x squared

Not $2x$. Check with a number: if $x = 5$, then $x^2 = 5 \times 5 = 25$ but $2x = 2 \times 5 = 10$. Different!

Opening the envelope

If someone tells you what number is inside, you can replace the letter with that number and work out the value. This is called substituting. Let's say $x = 4$.

$3x \longrightarrow 3 \times 4 \longrightarrow 12$ *3x means 3 times x, and x is 4*

$x^2 \longrightarrow 4 \times 4 \longrightarrow 16$ *x squared means x times x*

$2x + 3 \longrightarrow 2 \times 4 + 3 \longrightarrow 11$ *double 4 is 8, then add 3*

$\frac{x}{2} \longrightarrow \frac{4}{2} \longrightarrow 2$ *4 shared into 2 is 2*

The orange 4s show where the envelope was opened. Nothing else changed. That will be true all through this book: orange marks the part of a line that just changed.

Why use letters at all?

Because a letter lets you say something is true **for every number at once**. For example, try adding two numbers in either order:

$$2 + 5 = 5 + 2 \qquad 10 + 1 = 1 + 10 \qquad 7 + 30 = 30 + 7$$

You could write examples forever. Or you can write it once, with letters:

$$a + b = b + a$$

and that one short line says: "whatever two numbers you pick, adding them in either order gives the same answer". The letters are envelopes that can hold **any** number.

Round brackets $( )$ are a fence around part of a calculation. Whatever is inside the fence gets worked out **first**, and then treated as a single number.

Watch the same three numbers give different answers:

$$(2+3)\times 4$$

fence first: 2+3 is 5

$$\downarrow$$

$$5\times 4$$

$$\downarrow$$

$$20$$

$$2+3\times 4$$

$$\downarrow$$

no fence: times goes before plus

$$2+12$$

$$\downarrow$$

$$14$$

Without a fence there is an agreed order ("times and divide before plus and minus"). The full set of rules, and the famous internet argument about $8 \div 2(2+2)$, are in Chapter 1.

A number touching a bracket means multiply

4 times the result of (2 + 3) ..... $4(2+3)$

say: four, bracket, two plus three

There is an invisible times sign between the 4 and the bracket: $4(2+3) = 4 \times (2+3) = 4 \times 5 = 20$.

3 times the result of (x + 1) ..... $3(x+1)$

say: three, bracket, x plus one

Same idea with a letter. (How to "open" a bracket like this is in Chapter 11.)

A **formula** is a shorthand recipe that connects some quantities with letters. You will meet lots of them. Here is how to read them. Say each one out loud as you read.

$$y = mx + c$$

y — m times x — plus c

say: *"y equals m x plus c" (m and x are touching, so they are multiplied)*

$$v = u + at$$

v — u — plus a times t

say: *"v equals u plus a t" (this one is from physics: speed after some time)*

A Greek letter or two will turn up. The first is π, said "pie". It is not a variable: it is one fixed number, about 3.14159, that has to do with circles (Chapter 5 explains where it comes from).

$$A = \pi r^2$$

A — pi times — r squared

say: *"A equals pie r squared" (the area of a circle with radius r)*

Two more reading tips

Small letters written low, like x_1 and x_2, are just name tags. x_1 is said "x one" and means "the first x". It is **not** x times 1.

A letter with a little mark, like a' ("a dash" or "a prime") or $\bar{x}$ ("x bar"), is a different quantity related to a or x: $\bar{x}$ is the mean of the x values (Chapter 24), and $f'(x)$ is the gradient function of f (Chapter 34). Each mark is explained when it first appears.

Formulas with layers: read from the inside out

Some formulas are built in layers, like an onion. Your Lecture 2 slides use this one, for how long a pendulum (a swinging weight on a string) takes to swing:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Don't read it left to right. Find the innermost piece and work outwards:

layer 1 (the middle): $\frac{l}{g}$ *"l divided by g"*

layer 2: $\sqrt{\frac{l}{g}}$ *"the square root of that"*

layer 3: $2\pi\sqrt{\frac{l}{g}}$ *"two pi times that"*

outside: $T = 2\pi\sqrt{\frac{l}{g}}$ *"T is equal to all of that"*

So the whole thing is read: *"T equals two pi times the square root of l over g"*.

You now know how to read it. How to rearrange it, for example to find the length of string needed for a given swing time, is exactly what Chapter 14 teaches, one layer at a time.

Lesson 7 · How to read a page of maths

Maths is not read like a novel. A single line can take five minutes, and that is normal: people who are good at maths read it slowly too. Here is a route to follow whenever a line looks scary. Walk it one stepping stone at a time.

1

Say it out loud in English.

Use the phrases in this part. If you can say it, you can start to understand it.

2

Find the words you don't know.

The Word Finder at the back of the book lists every maths word with the page where it is properly explained.

3

Find the rule first.

Each section opens with a boxed RULE or PROCEDURE; read it before the example. Every chapter's rules are listed together on its rules page, and the Formula Pages at the back gather them all, with tables of standard derivatives and integrals.

4

Put real numbers in.

Replace each letter with a small, easy number such as 2 or 3 and work it out. Watching what happens teaches you more than staring.

5

Compare each line with the one above.

Ask: what ONE thing changed? Why was that allowed? In this book the change is marked and the reason is written beside it.

6

Cover the answer and try.

Hide the rest of a worked example with a sheet of paper and uncover one line at a time. Guess the next line before you look.

7

Stuck? Circle it and move on.

Mark the line in red on this page. Understanding often arrives on the second or third visit, after your brain has slept on it.

Your map of the book

The book is a railway network. Each coloured line is a Part; each stop is a chapter. Chapters only use ideas from stops earlier on the network, so if you get lost you can always travel back. Part 4 only needs Parts 1 and 2, so you can jump to it early.



Part 0 (this part) is the ticket office. Everything starts here.

Got a lecture coming up?

Each of your NTU maths lectures, and the chapters of this book that prepare you for it. The numbers in circles are chapter numbers, listed in the order to read them. The colour tells you which Part of the book (and which line on the railway map) they are in.

L01	Combining fractions and mathematical writing	read: 1 2 3
L02	Algebraic manipulation	read: 8 11 12 14
L03	Ratios and proportions	read: 4 15
L04	Data and descriptive statistics	read: 24
L05	Graphical analysis	read: 23
L06	Trigonometry	read: 27 28 29 30 31
L07	Powers, logs and graphing	read: 9 19 20 21 22
L08	Probability and Bayes	read: 25
L09	The Normal distribution	read: 26
L10	Vectors and scalars	read: 32
L11	Polar coordinates	read: 30 33
L12	Matrices and systems of equations	read: 15 16
L13	Functions and series	read: 6 17 18
L14	Differentiation and curves	read: 15 20 34
L15	Further differentiation	read: 35
L16	Introduction to integration	read: 36
L17	Further integration	read: 37

A note about your slides. A few slides contain small mistakes (typos in formulas). When a chapter meets one, it says so and shows the correct version. They are all collected in the Slide Errata at the back.